UNIFORMLY BOUNDED RADIAL VARIATION FOR THE $a_k = 1 - 1/k^2$ BLASCHKE PRODUCT AND RESOLUTION OF A PROBLEM ON T_g ACTING ON H^∞

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Abstract. Let $T_g f(z) = \int_0^z f(w)g'(w)\,dw$ and $S_g f(z) = \int_0^z f'(w)g(w)\,dw$ denote the generalized Volterra operator and its companion acting on $H^\infty(\mathbb{D})$. In [5, Problem 4.3], Anderson, Jovovic, and Smith asked for a zero-sequence characterization of Blaschke products belonging to the class $\mathrm{BRV} = \{g \in H(\mathbb{D}) : \sup_\theta \int_0^1 |g'(te^{i\theta})|\,dt < \infty\} \subseteq T[H^\infty] = S[H^\infty]$, singling out the Blaschke product $B(z)$ with zeros $a_k = 1 - 1/k^2$ as a critical borderline test case whose individual radii have rectifiable images, but for which a uniform arc-length bound was unknown. This sequence lies outside both previously known regimes: it is neither uniformly Frostman (Cargo's sufficient condition fails at $\theta = 0$) nor interpolating (Theorem 2.10 of [5] does not apply since $\rho(a_k, a_{k+1}) \sim 1/k \to 0$).

We resolve this problem in the affirmative by proving that for $a_k = 1 - 1/k^2$—and more generally for $a_k = 1 - 1/k^p$ for every $p > 1$—the Blaschke product $B(z)$ belongs to BRV, so T_B and S_B are bounded (and non-compact) operators on $H^\infty(\mathbb{D})$. The key analytic tool is a universal pole-free pointwise derivative inequality valid for every Blaschke product at every point $z \in \mathbb{D}$ (including at its zeros):

$$|B'(z)| \le e^{1/2} S(z) \exp\!\big(-\tfrac{1}{2}(1-|z|^2)S(z)\big), \qquad S(z) := \sum_k \frac{1-|a_k|^2}{|1-\bar{a}_k z|^2}.$$

Along each radius $z = te^{i\theta}$, the kernel $S(te^{i\theta}) \asymp \big((1-t) + 2|\sin(\theta/2)|\big)^{-(1+1/p)}$ induces a two-zone radial decomposition on which the integrands admit exact elementary exponential antiderivatives whose increments are bounded independently of θ.

1. Introduction and Operator-Theoretic Background

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ be the open unit disk, $H(\mathbb{D})$ the space of holomorphic functions on $\mathbb{D}$, $H^\infty = H^\infty(\mathbb{D})$ the Banach algebra of bounded holomorphic functions with norm $\|f\|_\infty = \sup_{z\in\mathbb{D}} |f(z)|$, and $A = A(\mathbb{D})$ the disk algebra of functions in H^∞ extending continuously to $\overline{\mathbb{D}}$. For a symbol $g \in H(\mathbb{D})$, the Volterra-type integral operator T_g, its companion operator S_g, and the multiplication operator M_g are defined for $f \in H(\mathbb{D})$ and $z \in \mathbb{D}$ by

$$T_g f(z) = \int_0^z f(w)g'(w)\,dw, \qquad S_g f(z) = \int_0^z f'(w)g(w)\,dw, \qquad M_g f(z) = f(z)g(z). \tag{1.1}$$

Integration by parts connects the three operators via the fundamental identity [5, Eq. (1.1)]

$$M_g f(z) = f(0)g(0) + T_g f(z) + S_g f(z). \tag{1.2}$$

For a Banach space $X \subseteq H(\mathbb{D})$ and $L \in \{T, S, M\}$, write $L[X] = \{g \in H(\mathbb{D}) : L_g \text{ is bounded on } X\}$ and $L_0[X] = \{g \in H(\mathbb{D}) : L_g \text{ is compact on } X\}$. On classical Hardy, Bergman, Dirichlet, Bloch,

Date: November 5, 2026.

2020 *Mathematics Subject Classification.* Primary: 30J10, 47B38; Secondary: 30H05, 68V15.

Key words and phrases. Blaschke products, bounded radial variation, Volterra-type integral operators, H^∞, Frostman condition, interpolating sequences, Lean 4 formal verification.

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and BMOA spaces, boundedness of M_g splits into boundedness of S_g (which is equivalent to $g \in H^\infty$) and boundedness of T_g (which is governed by a Carleson or Bloch-type condition on g'); see [1, 2, 3, 4, 9, 10]. On H^∞, however, $M[H^\infty] = H^\infty$ trivially, and by [5, Proposition 2.3] the identity (1.2) forces

$$T[H^\infty] = S[H^\infty] \subsetneq M[H^\infty] = H^\infty, \tag{1.3}$$

so $T[H^\infty]$ is a non-closed subalgebra of H^∞ ([5, Corollary 2.15]).

For $\theta \in [-\pi, \pi]$, the *radial variation* of $g \in H(\mathbb{D})$ along the radius $e^{i\theta}$ is

$$V(g, \theta) := \int_0^1 |g'(te^{i\theta})|\, dt, \tag{1.4}$$

and the class of analytic functions with *uniformly bounded radial variation* is

$$\mathrm{BRV} := \Big\{ g \in H(\mathbb{D}) : \|g\|_{\mathrm{BRV}} := \sup_{\theta \in [-\pi,\pi]} V(g,\theta) < \infty \Big\}. \tag{1.5}$$

Since $|T_g f(re^{i\theta})| = \left|\int_0^r f(te^{i\theta}) g'(te^{i\theta}) e^{i\theta}\, dt\right| \le V(g,\theta)\|f\|_\infty$, one has $\mathrm{BRV} \subseteq T[H^\infty] = S[H^\infty]$ with $\|T_g\|_{H^\infty \to H^\infty} \le \|g\|_{\mathrm{BRV}}$ and $\|S_g\|_{H^\infty \to H^\infty} \le 2\|g\|_\infty + \|g\|_{\mathrm{BRV}}$ ([5, Proposition 2.4]). Whether $\mathrm{BRV} = T[H^\infty]$ remains Open Problem 4.1 of [5].

1.1. **Blaschke Products in** BRV **and Problem 4.3 of** [5]**.** A Blaschke product with zero sequence $\{a_k\} \subset \mathbb{D} \setminus \{0\}$ satisfying $\sum_k (1 - |a_k|) < \infty$ is given by

$$B(z) = \prod_k \frac{|a_k|}{a_k} \frac{a_k - z}{1 - \bar{a}_k z}, \qquad z \in \mathbb{D}.$$

Prior to the present work, two contrasting regimes of Blaschke products were understood in [5]:

(1) **Uniformly Frostman Blaschke Products ($B \in$ BRV):** A Blaschke product B is called *uniformly Frostman* [5, Eq. (2.2)] if there exists a constant $C > 0$ independent of θ such that

$$\sup_{\theta \in [-\pi,\pi]} \sum_k \frac{1 - |a_k|}{|a_k - e^{i\theta}|} < C. \tag{1.6}$$

By Cargo's theorem [6, Theorem 1], B is uniformly Frostman if and only if *every subproduct* of B belongs to BRV. However, condition (1.6) forces $\{a_k\}$ to approach $\partial\mathbb{D}$ *tangentially*: no infinite sequence $\{a_k\}$ contained in a nontangential Stolz angle (and in particular no real zero sequence $0 < a_k < 1$) can satisfy (1.6).

(2) **Real Interpolating Blaschke Products ($B \notin T[H^\infty]$):** Recall that $\{a_k\}$ is an *interpolating sequence* for H^∞ if $\inf_k \prod_{j \ne k} \rho(a_j, a_k) \ge \delta > 0$, where $\rho(z,w) = \left|\frac{z-w}{1-\bar{w}z}\right|$ is the pseudohyperbolic metric. By [5, Proposition 2.7 and Theorem 2.10], if B is an interpolating Blaschke product with zeros in a nontangential region, then $B \notin$ BRV; moreover, if $0 < a_k < a_{k+1} < 1$ is an interpolating sequence on $(0,1)$, then T_B is *unbounded* on H^∞ ($B \notin T[H^\infty]$).

This dichotomy led Anderson, Jovovic, and Smith to pose [5, Problem 4.3, p. 14]:

Problem 1.1 ([5, Problem 4.3])**.** *Characterize, in terms of their zero sequences, Blaschke products in* BRV*. A possible candidate for a Blaschke product in* BRV *is the Blaschke product with zeros* $\{1 - 1/k^2\}$*. It is not hard to see that the image of every radius is rectifiable, but it is not clear that there is a uniform bound for their arc-lengths.*

1.2. **Main Results.** In this paper, we resolve the candidate of Problem 1.1 in the affirmative and extend the result to the entire family of polynomial-clustering real Blaschke sequences $a_k = 1 - 1/k^p$ for all $p > 1$.

Theorem 1.2 (Main Theorem: Resolution of the Candidate in Problem 1.1)**.** *For every $p > 1$, let $a_1 = 0$ and $a_k = 1 - 1/k^p \in (0,1)$ for $k \geq 2$, and let*

$$B(z) = -z \prod_{k=2}^{\infty} \frac{a_k - z}{1 - a_k z}, \qquad z \in \mathbb{D}, \tag{1.7}$$

be the corresponding Blaschke product (in particular, $p = 2$ gives $a_k = 1 - 1/k^2$). Then:

(1) *B is **not** uniformly Frostman (so its interpolating subproducts do not belong to* BRV*), and $\{a_k\}$ is **not** an interpolating sequence ($\rho(a_k, a_{k+1}) \sim \frac{p}{2k} \to 0$).*
(2) *B has **uniformly bounded radial variation** on $\mathbb{D}$:*

$$\|B\|_{\mathrm{BRV}} = \sup_{\theta \in [-\pi,\pi]} \int_0^1 |B'(te^{i\theta})|\, dt < \infty, \tag{1.8}$$

and hence $B \in \mathrm{BRV} \subseteq T[H^\infty] = S[H^\infty]$.
(3) *Both T_B and S_B are bounded linear operators on $H^\infty(\mathbb{D})$, whereas $B \notin A(\mathbb{D})$ and therefore $B \notin T_0[H^\infty]$ (T_B is bounded but **not compact** on $H^\infty(\mathbb{D})$, and $B' \notin \mathcal{U}$).*

The mechanism behind Theorem 1.2 reveals a striking analytic phenomenon. An individual Blaschke product with real zeros $a_k \to 1^-$ can belong to BRV precisely when its zeros cluster tightly enough in the pseudohyperbolic metric that the exponential attenuation of $|B(z)|$ near $z = 1$ crushes the blowup of the logarithmic derivative $|B'(z)/B(z)|$. We capture this cancellation globally on $\mathbb{D}$ through a universal pole-free derivative bound (Theorem 3.1) and a two-zone radial scale decomposition with exact elementary antiderivatives (Section 4).

2. Borderline Geometry and Real-Ray Variation of $a_k = 1 - 1/k^2$

By the Leibniz rule $(g_1 g_2)' = g_1' g_2 + g_1 g_2'$, any two functions $g_1, g_2 \in \mathrm{BRV} \cap H^\infty$ satisfy

$$V(g_1 g_2, \theta) \leq \|g_2\|_\infty V(g_1, \theta) + \|g_1\|_\infty V(g_2, \theta). \tag{2.1}$$

In particular, since the origin factor $g_1(z) = -z$ satisfies $\|g_1\|_\infty = 1$ and $V(g_1, \theta) = 1$, removing the single factor $-z$ at $k = 1$ changes $V(B, \theta)$ by at most 1. Throughout the remainder of the paper we therefore work with the positive-zero Blaschke product

$$B(z) = \prod_{k=2}^{\infty} b_k(z), \qquad b_k(z) := \frac{a_k - z}{1 - a_k z}, \qquad a_k = 1 - \frac{1}{k^2} \in \left[\tfrac{3}{4}, 1\right). \tag{2.2}$$

Proposition 2.1 (Blaschke Convergence, Non-Frostman Property, and Non-Separation)**.** *Let $a_k = 1 - 1/k^2$ for $k \geq 2$. Then:*

(1) ***Blaschke Condition:** For every $k \geq 2$, $1 - a_k = \frac{1}{k^2} \leq \frac{1}{k-1} - \frac{1}{k}$, and $\frac{7}{4k^2} \leq 1 - a_k^2 \leq \frac{2}{k^2}$. Hence $\sum_{k=2}^{\infty}(1 - a_k) = \frac{\pi^2}{6} - 1 < 1$.*
(2) ***Failure of Uniform Frostman Condition at $\theta = 0$:** At $e^{i0} = 1$, every term of the Frostman sum (1.6) equals $\frac{1-a_k}{|a_k - 1|} = 1$, so for every $N \geq 1$,*

$$\sum_{k=2}^{N+1} \frac{1 - a_k}{|a_k - 1|} = N \longrightarrow \infty \qquad (N \to \infty). \tag{2.3}$$

(3) ***Failure of Pseudohyperbolic Separation:*** *For every $k \geq 1$, the pseudohyperbolic distance between consecutive zeros satisfies the exact identity*

$$\rho(a_k, a_{k+1}) = \frac{a_{k+1} - a_k}{1 - a_k a_{k+1}} = \frac{2k+1}{2k^2+2k} \leq \frac{1}{k} \longrightarrow 0 \qquad (k \to \infty). \tag{2.4}$$

Consequently, $\{a_k\}$ is not separated in the pseudohyperbolic metric and is not an interpolating sequence for H^∞.

Proof. (1) Since $k(k-1) < k^2$, we have $\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$. For $k \geq 2$, $\frac{3}{4} \leq a_k < 1$ gives $\frac{7}{4} \leq 1 + a_k < 2$, so $\frac{7}{4k^2} \leq (1-a_k)(1+a_k) = 1 - a_k^2 < \frac{2}{k^2}$.

(2) Since $a_k < 1$, $|a_k - 1| = 1 - a_k > 0$, so each ratio in (2.3) is 1.

(3) Direct algebraic simplification gives $a_{k+1} - a_k = \frac{1}{k^2} - \frac{1}{(k+1)^2} = \frac{2k+1}{k^2(k+1)^2}$ and

$$1 - a_k a_{k+1} = 1 - \left(1 - \frac{1}{k^2}\right)\left(1 - \frac{1}{(k+1)^2}\right) = \frac{(k+1)^2 + k^2 - 1}{k^2(k+1)^2} = \frac{2k^2+2k}{k^2(k+1)^2}.$$

Dividing the two expressions cancels $k^2(k+1)^2$ and yields $\frac{2k+1}{2k^2+2k} \leq \frac{2k+2}{2k^2+2k} = \frac{1}{k}$. □

Before proving the general uniform bound for all $\theta \in [-\pi, \pi]$ in Sections 3–4, it is instructive to see directly on the positive real axis $\theta = 0$ why $V(B, 0) = \int_0^1 |B'(t)|\, dt < \infty$, and how the critical-point interlacing of [5, Lemma 2.9] converts pseudohyperbolic clustering into exponential decay between zeros.

Theorem 2.2 (Real-Ray Variation via Critical-Point Interlacing)**.** *Along the positive real radius $\theta = 0$, the Blaschke product $B(z) = \prod_{k=2}^{\infty} \frac{a_k - z}{1 - a_k z}$ with $a_k = 1 - 1/k^2$ satisfies*

$$V(B, 0) = \int_0^1 |B'(t)|\, dt \leq 1 + 2\sum_{k=2}^{\infty} e^{-k/128} = 1 + \frac{2e^{-1/64}}{1 - e^{-1/128}} < \infty. \tag{2.5}$$

Proof. By [5, Lemma 2.9] (proved via Rolle's theorem and Hurwitz's theorem), the derivative B' is real-valued on $(0, 1)$, has no zeros on $[0, a_2)$, and has a unique zero $c_k \in (a_k, a_{k+1})$ in each interval (a_k, a_{k+1}) for $k \geq 2$. Since $B(a_k) = 0$, B is monotone on $[0, a_2]$, $[a_k, c_k]$, and $[c_k, a_{k+1}]$, so

$$\int_0^{a_2} |B'(t)|\, dt = |B(a_2) - B(0)| = B(0) \leq 1, \qquad \int_{a_k}^{a_{k+1}} |B'(t)|\, dt = 2|B(c_k)| \leq 2 \sup_{t \in [a_k, a_{k+1}]} |B(t)|.$$

For any $t \in [a_k, a_{k+1}]$ and any index $j \in \{k+1, \ldots, 2k\}$, we have $1 - a_j^2 \geq 1 - a_j = \frac{1}{j^2} \geq \frac{1}{4k^2}$, $1 - t^2 \geq 1 - t \geq 1 - a_{k+1} = \frac{1}{(k+1)^2} \geq \frac{1}{4k^2}$, and $0 < 1 - a_j t = (1 - a_j) + a_j(1 - t) \leq \frac{1}{k^2} + \frac{1}{k^2} = \frac{2}{k^2}$. Hence

$$1 - |b_j(t)|^2 = \frac{(1 - a_j^2)(1 - t^2)}{(1 - a_j t)^2} \geq \frac{(1/4k^2)(1/4k^2)}{(2/k^2)^2} = \frac{1}{64}.$$

Using $|b_j(t)| = \sqrt{1 - (1 - |b_j(t)|^2)} \leq e^{-(1 - |b_j(t)|^2)/2} \leq e^{-1/128}$ across the k factors $j = k+1, \ldots, 2k$ gives $|B(c_k)| \leq \prod_{j=k+1}^{2k} |b_j(c_k)| \leq e^{-k/128}$, and summing over $k \geq 2$ yields (2.5). □

3. The Universal Pole-Free Pointwise Blaschke Derivative Bound

On off-real rays $\theta \neq 0$, Rolle's theorem is unavailable, and the naive logarithmic derivative formula $B'(z) = B(z) \sum_k \frac{-(1-a_k^2)}{(a_k - z)(1 - a_k z)}$ has apparent poles at the zeros $z = a_k$. Our key observation is that differentiating B via the omitted-subproduct expansion $B_k(z) = \prod_{j \neq k} b_j(z)$ eliminates the $(a_k - z)^{-1}$ singularity altogether, while losing at most a factor of $e^{1/2}$ from the single omitted factor $b_k(z)$.

Theorem 3.1 (Universal Pole-Free Blaschke Derivative Inequality)**.** *Let $B(z)=\prod_k \frac{|a_k|}{a_k}\frac{a_k-z}{1-\bar a_k z}$ be any Blaschke product on $\mathbb{D}$, and define its positive kernel function*

$$S(z):=\sum_k \frac{1-|a_k|^2}{|1-\bar a_k z|^2},\qquad z\in\mathbb{D}. \tag{3.1}$$

*Then at **every** point $z\in\mathbb{D}$ (including at all zeros $z=a_k$),*

$$|B'(z)|\le e^{1/2}\,S(z)\,\exp\!\left(-\frac{1}{2}(1-|z|^2)S(z)\right)\le e^{1/2}\,S(z)\,\exp\!\left(-\frac{1}{2}(1-|z|)S(z)\right). \tag{3.2}$$

Proof. Let $b_j(z)=\frac{|a_j|}{a_j}\frac{a_j-z}{1-\bar a_j z}$ and $B_k(z)=\prod_{j\neq k} b_j(z)$. Since $\frac{d}{dz}b_k(z)=-\frac{|a_k|}{a_k}\frac{1-|a_k|^2}{(1-\bar a_k z)^2}$, the Leibniz rule on the locally uniformly convergent product $B=\prod_j b_j$ gives

$$B'(z)=-\sum_k \frac{|a_k|}{a_k}B_k(z)\frac{1-|a_k|^2}{(1-\bar a_k z)^2}. \tag{3.3}$$

For each factor $b_j(z)$, the elementary identity $|1-\bar a_j z|^2-|a_j-z|^2=(1-|a_j|^2)(1-|z|^2)$ yields

$$u_j(z):=1-|b_j(z)|^2=\frac{(1-|a_j|^2)(1-|z|^2)}{|1-\bar a_j z|^2}\in[0,1]. \tag{3.4}$$

Since $1-u\le e^{-u}$ for all $u\in\mathbb{R}$, we have $|b_j(z)|=\big(1-u_j(z)\big)^{1/2}\le\exp\big(-\frac{1}{2}u_j(z)\big)$. For any fixed index k, the single omitted term satisfies $u_k(z)=1-|b_k(z)|^2\le 1$, and $\sum_j u_j(z)=(1-|z|^2)S(z)$. Therefore, for every k and every $z\in\mathbb{D}$,

$$\begin{aligned}|B_k(z)|=\prod_{j\neq k}|b_j(z)|&\le\exp\left(-\frac{1}{2}\sum_{j\neq k}u_j(z)\right)=\exp\!\left(-\frac{1}{2}\big((1-|z|^2)S(z)-u_k(z)\big)\right)\\&\le e^{1/2}\exp\!\left(-\frac{1}{2}(1-|z|^2)S(z)\right).\end{aligned} \tag{3.5}$$

Substituting (3.5) into the triangle inequality for (3.3) and using $1-|z|^2=(1+|z|)(1-|z|)\ge 1-|z|$ gives (3.2). □

3.1. **Two-Sided Power Law for the Kernel $S(te^{i\theta})$.** We now specialize to $a_k=1-1/k^p$ ($p>1$, $k\ge 2$) along any radius $z=te^{i\theta}$ ($t\in[0,1)$, $\theta\in[-\pi,\pi]$). Write $x:=1-t\in(0,1]$, $s:=2|\sin(\theta/2)|\in[0,2]$, and define the combined radial-angular scale parameter

$$\delta=\delta(t,\theta):=(1-t)+2|\sin(\theta/2)|=x+s. \tag{3.6}$$

Lemma 3.2 (Two-Sided Denominator Scale Bound)**.** *For every $a_k\in[3/4,1)$, every $t=1-x\in[1/2,1)$ ($x\in(0,1/2]$), and every $\theta\in[-\pi,\pi]$ with $s=2|\sin(\theta/2)|$ and $\delta=x+s$,*

$$|1-a_k te^{i\theta}|^2=(1-a_k t)^2+4a_k t\sin^2(\theta/2)=(1-a_k t)^2+a_k t s^2, \tag{3.7}$$

and consequently

$$\frac{3}{16}\big((1-a_k)+\delta\big)^2\le|1-a_k te^{i\theta}|^2\le\big((1-a_k)+\delta\big)^2. \tag{3.8}$$

Proof. Expanding $|1-a_k te^{i\theta}|^2=1-2a_k t\cos\theta+a_k^2t^2$ and substituting $\cos\theta=1-2\sin^2(\theta/2)$ proves (3.7). Since $1-a_k t=(1-a_k)+a_k x$ with $\frac{3}{4}\le a_k<1$ and $\frac{1}{2}\le t<1$, we have $\frac{3}{4}((1-a_k)+x)\le 1-a_k t\le(1-a_k)+x$ and $\frac{3}{8}\le a_k t<1$. For the upper bound,

$$(1-a_k t)^2+a_k t s^2\le\big((1-a_k)+x\big)^2+s^2\le\big((1-a_k)+x+s\big)^2=\big((1-a_k)+\delta\big)^2.$$

For the lower bound, using $A^2+B^2 \geq \frac{1}{2}(A+B)^2$ with $A=(1-a_k)+x$ and $B=s$ gives

$$(1-a_kt)^2+a_kts^2 \geq \frac{9}{16}\big((1-a_k)+x\big)^2+\frac{3}{8}s^2 \geq \frac{3}{8}\Big(\big((1-a_k)+x\big)^2+s^2\Big) \geq \frac{3}{16}\big((1-a_k)+\delta\big)^2. \ \square$$

Proposition 3.3 (Two-Sided Power Law for $S(te^{i\theta})$)**.** *Let $a_k = 1-1/k^p$ for $k \geq 2$ with $p>1$ (and $a_k \geq 3/4$ for $k \geq k_0$). There exist constants $C_1, C_2 > 0$ depending only on p such that for all $t \in [1/2,1)$ and $\theta \in [-\pi,\pi]$ with $\delta = (1-t)+2|\sin(\theta/2)| \leq 2^{-p}$:*

$$C_1\,\delta^{-(1+1/p)} \leq S(te^{i\theta}) \leq C_2\,\delta^{-(1+1/p)}. \tag{3.9}$$

In particular, for $p=2$ ($a_k = 1-1/k^2$, $k_0=2$ and $\delta \leq 1/4$), (3.9) holds with the explicit constants $C_1 = \frac{7}{192}$ and $C_2 = 32$.

Proof. By Lemma 3.2 and $\frac{7}{4}k^{-p} \leq 1-a_k^2 \leq 2k^{-p}$, we have

$$\frac{7}{4}\Sigma_p(\delta) \leq S(te^{i\theta}) \leq \frac{32}{3}\Sigma_p(\delta), \qquad \text{where } \Sigma_p(\delta) := \sum_{k=2}^{\infty}\frac{k^{-p}}{(k^{-p}+\delta)^2} = \sum_{k=2}^{\infty}\frac{k^p}{(1+\delta k^p)^2}.$$

Let $N := \lfloor \delta^{-1/p} \rfloor \geq 2$.

- *Lower bound:* For every $k \in \{2,\dots,N\}$, we have $\delta k^p \leq 1$, so $(1+\delta k^p)^2 \leq 4$ and

$$\Sigma_p(\delta) \geq \frac{1}{4}\sum_{k=2}^{N} k^p \geq \frac{1}{4}\int_1^N u^p\,du = \frac{N^{p+1}-1}{4(p+1)} \geq c_1(p)\delta^{-(1+1/p)}.$$

  For $p=2$ and $N \geq 2$, $\frac{1}{4}\sum_{k=2}^{N}k^2 = \frac{N(N+1)(2N+1)-6}{24} \geq \frac{N^3}{12} \geq \frac{(\delta^{-1/2}/2)^3}{12} = \frac{1}{96}\delta^{-3/2}$, giving $C_1 = \frac{7}{4}\cdot\frac{1}{96} < \frac{7}{192}$.
- *Upper bound:* Splitting $\Sigma_p(\delta)$ at $N = \lfloor \delta^{-1/p} \rfloor$ gives $\sum_{k=2}^{N}\frac{k^p}{(1+\delta k^p)^2} \leq \sum_{k=2}^{N}k^p \leq N\cdot N^p \leq \delta^{-(1+1/p)}$ on the bulk $k \leq N$, while on the tail $k > N$, $1+\delta k^p > \delta k^p$ gives

$$\sum_{k=N+1}^{\infty}\frac{k^p}{(1+\delta k^p)^2} \leq \frac{1}{\delta^2}\sum_{k=N+1}^{\infty}k^{-p} \leq \frac{1}{\delta^2}\int_N^{\infty}u^{-p}\,du = \frac{N^{1-p}}{(p-1)\delta^2} \leq \frac{2^{p-1}}{p-1}\delta^{-(1+1/p)}.$$

  For $p=2$, $1+\frac{2^{2-1}}{2-1}=3$, so $S(te^{i\theta}) \leq \frac{32}{3}\cdot 3\delta^{-3/2} = 32\delta^{-3/2}$. $\square$

4. Exact Elementary Antiderivatives and Proof of Theorem 1.2

We now combine Theorem 3.1 and Proposition 3.3 to bound $V(B,\theta) = \int_0^1 |B'(te^{i\theta})|\,dt$ uniformly in $\theta \in [-\pi,\pi]$.

Proof of Theorem 1.2. Fix $p>1$, let $\alpha := 1+1/p \in (1,2)$, and let $x_0 := \frac{1}{2}2^{-p} \in (0,1/4]$ (so for $p=2$, $\alpha = 3/2$ and $x_0 = 1/8$). For any $\theta \in [-\pi,\pi]$, set $s = 2|\sin(\theta/2)| \geq 0$ and change variables to $x = 1-t \in (0,1]$.

Case 1 ($s \geq x_0$): When $s \geq x_0$, every $t \in [0,1)$ satisfies $|1-a_kte^{i\theta}| \geq \max(1-t, a_2ts/2) \geq c_0(p) > 0$, so $S(te^{i\theta}) \leq c_0(p)^{-2}\sum_{k=2}^{\infty}(1-a_k^2) < \infty$ is uniformly bounded on $[0,1)$.

Case 2 ($0 \leq s < x_0$): Split $[0,1]$ into three radial zones in $x = 1-t$:

(1) **Zone III (Compact Core, $x \in [x_0,1]$, i.e. $t \in [0,1-x_0]$):** Here $|1-a_kz| \geq 1-|z| = x \geq x_0$, so

$$\int_{x_0}^1 |B'((1-x)e^{i\theta})|\,dx \leq \int_{x_0}^1 S((1-x)e^{i\theta})\,dx \leq \frac{1}{x_0^2}\sum_{k=2}^{\infty}(1-a_k^2) =: M_{\text{core}} < \infty. \tag{4.1}$$

(2) **Zone I (Outer / Boundary Layer, $0 < x \le s$):** Here $\delta = x + s \in [s, 2s] \subseteq (0, 2x_0]$, so Proposition 3.3 gives $S \le C_2 s^{-\alpha}$ and $S \ge C_1 (2s)^{-\alpha} = 2^{-\alpha} C_1 s^{-\alpha}$. Setting $c' := \frac{1}{2} 2^{-\alpha} C_1 > 0$ and $\Lambda := c' s^{-\alpha} > 0$, Theorem 3.1 yields

$$|B'((1-x)e^{i\theta})| \le e^{1/2} C_2 \, s^{-\alpha} \exp\bigl(-c' s^{-\alpha} x\bigr) = \frac{d}{dx}\left(-\frac{e^{1/2} C_2}{c'} \exp(-\Lambda x) \right).$$

Integrating this exact derivative over $x \in [0, s]$ gives the θ-independent bound

$$\int_0^s |B'((1-x)e^{i\theta})| \, dx \le \frac{e^{1/2} C_2}{c'} \bigl(1 - e^{-c' s^{1-\alpha}}\bigr) \le \frac{e^{1/2} C_2}{c'} < \infty. \tag{4.2}$$

(3) **Zone II (Inner / Bulk Critical Zone, $s \le x \le x_0$):** Here $\delta = x + s \in [x, 2x] \subseteq (0, 2x_0]$, so Proposition 3.3 gives

$$S((1-x)e^{i\theta}) \le C_2 x^{-\alpha} = C_2 x^{-(1+1/p)}, \qquad \frac{1}{2} x S((1-x)e^{i\theta}) \ge c' x^{1-\alpha} = c' x^{-1/p}.$$

Crucially, the derivative of the exponent $u(x) := -c' x^{-1/p}$ is $u'(x) = \frac{c'}{p} x^{-(1+1/p)}$, which matches the prefactor power $x^{-(1+1/p)}$ *identically*! Therefore, by Theorem 3.1,

$$|B'((1-x)e^{i\theta})| \le e^{1/2} C_2 \, x^{-(1+1/p)} \exp\Bigl(-c' x^{-1/p}\Bigr) = \frac{d}{dx}\left(\frac{p \, e^{1/2} C_2}{c'} \exp\Bigl(-c' x^{-1/p}\Bigr) \right).$$

Integrating this exact derivative over $x \in [s, x_0]$ gives

$$\int_s^{x_0} |B'((1-x)e^{i\theta})| \, dx \le \frac{p \, e^{1/2} C_2}{c'} \Bigl(e^{-c' x_0^{-1/p}} - e^{-c' s^{-1/p}} \Bigr) \le \frac{p \, e^{1/2} C_2}{c'} < \infty. \tag{4.3}$$

Summing (4.1), (4.2), and (4.3) proves that for all $\theta \in [-\pi, \pi]$,

$$V(B, \theta) \le \frac{(1+p) e^{1/2} C_2}{c'} + M_{\text{core}} < \infty, \tag{4.4}$$

so $B \in \mathrm{BRV} \subseteq T[H^\infty] = S[H^\infty]$. Finally, since $B(a_k) = 0$ with $a_k \to 1$ while $\lim_{r \to 1^-} |B(re^{i\theta})| = 1$ for all $\theta \in (-\pi, \pi) \setminus \{0\}$, B has no continuous extension to $z = 1$, so $B \notin A(\mathbb{D})$. By [5, Theorem 3.5], $T_0[H^\infty] \subsetneq A(\mathbb{D})$, whence $B \notin T_0[H^\infty]$ and T_B is not compact on $H^\infty(\mathbb{D})$. □

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